

ALFA RIZI

BACKEND DEVELOPER / JAVA ENGINEER

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PROFESSIONAL SUMMARY

Backend-focused Software Engineering student (GPA 3.6/4.0) with proven experience building distributed microservices, RESTful APIs, and database-driven architectures. Skilled in Java (Spring Boot 3, Spring Security, Spring Cloud Gateway), Python (FastAPI), and SQL (PostgreSQL, PL/pgSQL triggers, views, Flyway migrations). Experienced in containerization with Docker, JWT/RBAC security implementations, and automated CI/CD pipelines.

TECHNICAL SKILLS

Languages: Java 17, Python 3.12, SQL, JavaScript (Node.js), C
Frameworks & Microservices: Spring Boot 3.2, Spring Cloud Gateway, Spring Security 6, Spring Data JPA, FastAPI, Express.js
Databases & Migrations: PostgreSQL, PL/pgSQL (Triggers, Views, Functions), Flyway, Drizzle ORM, Hibernate, ChromaDB, SQLite
Security & Auth: JWT (Access & Refresh Tokens), RBAC, BCrypt, Rate Limiting, CORS, Helmet
DevOps & Tooling: Docker, Docker Compose, GitHub Actions (CI/CD), Git, Maven, Postman, Linux CLI, OpenAPI/Swagger

FEATURED BACKEND PROJECTS

ApexGrid — F1 Race Ticket Booking Backend | Java 17, Spring Boot, Spring Cloud Gateway, PostgreSQL, Docker 2026

- Architected a 3-tier Spring Boot microservices backend comprising an API Gateway, User Service, and Race Ticket Service via Spring Cloud Gateway.
- Secured inter-service routing and REST endpoints using JWT authentication, refresh token rotation, and Spring Security RBAC (USER/ADMIN).
- Engineered concurrency-safe ticket inventory management via custom PostgreSQL PL/pgSQL triggers (update_race_quota) and Spring @Scheduled workers.
- Developed automated PDF ticket generation with embedded QR codes (ZXing & iText) and asynchronous email notification tracking.
- Containerized the gateway, services, and database using Docker Compose with robust health checks and network isolation.

StudyMate AI — Asynchronous RAG Backend Engine | Python, FastAPI, ChromaDB, Sentence-Transformers, PyMuPDF 2026

- Designed an asynchronous document retrieval and study assistant API using FastAPI, PyMuPDF, and LangChain text splitters.
- Implemented semantic vector similarity search by embedding document chunks with Sentence Transformers (all-MiniLM-L6-v2) into ChromaDB.
- Connected an OpenAI-compatible LLM gateway to generate grounded study materials with verified [Page X] citations.
- Built a token-bucket IP rate limiter (60 req/min) and configured a GitHub Actions CI/CD workflow running Ruff, Pytest async, and pip-audit.

Finance Management System (FinanceAI) — Spring Boot MVC | Java 17, Spring Boot, Spring Security, PostgreSQL, Flyway 2026

- Engineered a robust personal finance web application using Spring Boot 3.2, Spring Data JPA, and PostgreSQL.
- Implemented complete database version control and schema evolution using Flyway migrations (V1__init_schema.sql).
- Configured Spring Security 6 with session-based authentication, BCrypt encryption, and role-based endpoint authorization.
- Created service layers calculating debt payoff schedules, emergency fund milestones, and financial health scores.

Finance Feasibility — REST API & Financial Modeling Backend | Node.js, Express.js, TypeScript, PostgreSQL, Drizzle ORM 2026

- Demo: <https://finance-feasibility.alfarizi.my.id> | Code: github.com/a1fariz/finance-feasibility
- Developed a high-performance REST API using Express.js and TypeScript, bundled with esbuild and managed via Drizzle ORM on PostgreSQL.
 - Implemented financial valuation algorithms for automated Net Present Value (NPV), IRR, ROI, and Payback Period computation.
 - Integrated Firebase Admin token verification middleware and connected Google Gemini API for strategic report generation.

PROFESSIONAL EXPERIENCE

Lead Intern — IT & Business Operations 2025 – Present

PT Resik Cemerlang (Pasim Group) Bandung, Indonesia

- Directed a 5-member intern team handling IT operations, troubleshooting system failures, and maintaining hardware/software SLAs.
- Maintained operating system stability and network security across company laptops via automated patching and antivirus deployment.
- Validated technical specifications across thousands of medical device data records to ensure database integrity.
- Delivered structured weekly incident and operational health reports to senior management.

Lead Intern — IT Hardware 2024

SMP Negeri 2 Batujajar West Bandung, Indonesia

- Conducted hardware fault diagnosis, network cabling repairs, and component servicing for school lab computers.
- Established an inventory tracking and maintenance log system that streamlined recurrent hardware issue resolutions.

EDUCATION & TRAINING

Universitas Nasional PASIM Expected May 2027

Associate Degree in Informatics Management (D3) — GPA: 3.6 / 4.0 Bandung, Indonesia

- PUB Intensive Technical Scholarship (2024 – Present): Algorithms, Data Structures, Relational Database Engineering, Advanced Java Development.